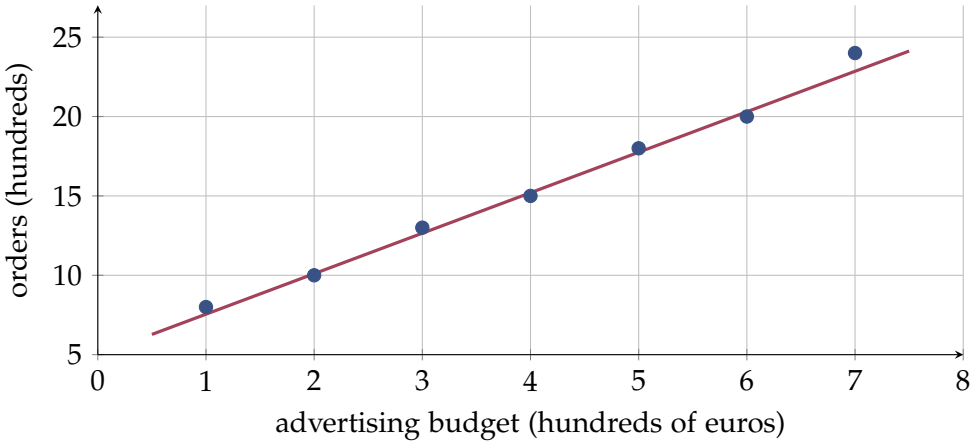
CHAPTER 5

Bivariate statistics

Skills: represent, choose a linear regression, interpolate, extrapolate with caution, communicate

5.1 Reference course

A bivariate data set consists of pairs (x_i, y_i). The scatter plot shows a trend, the spread and any outliers.



Key point. A line of best fit $y = ax + b$ gives estimates, not exact equalities. The coefficient a describes the estimated average change in y when x increases by one unit. Interpolation stays within the range of observed values; extrapolation goes outside that range and must be presented with caution.

The mean point $G(\bar{x}, \bar{y})$ has coordinates $\bar{x} = \dfrac{x_1 + \cdots + x_n}{n}$ and $\bar{y} = \dfrac{y_1 + \cdots + y_n}{n}$. A regression line obtained with a calculator passes through G, up to rounding.

Method. To use $y = ax + b$: 1) replace x by the requested value to estimate y; 2) solve $ax + b = y_0$ to estimate x; 3) round according to the context; 4) state whether it is interpolation or extrapolation; 5) remind that the result depends on the model.

A change of variable, for example $z = \log(y)$ or $z = \dfrac{1}{y}$, can make a scatter plot roughly linear. We then fit (x, z), and afterwards return to the variable y.

Session 31 — Scatter plot

Bivariate statistics

26 Jan 2027 • B: 2 h – C: 1 h

Objectives.

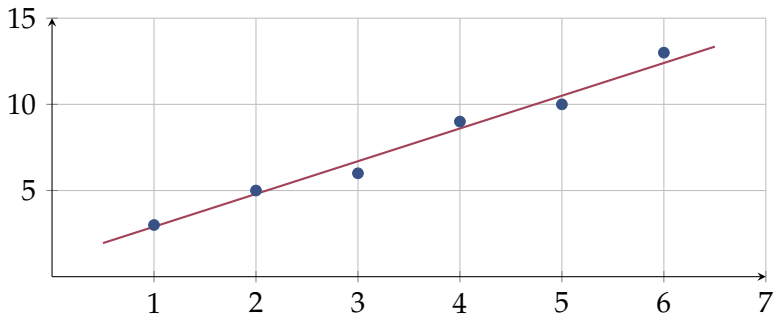
- Construct a scatter plot.
- Choose the scales.
- Identify the variables.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 10% increase. $1 + \dfrac{10}{100} = 1.10$
2. Solve $3x + 4 = 25$. $x = 7$
3. Calculate the mean of 6, 8 and 10. 8
4. Give the slope between $(1,3)$ and $(4,9)$. $\dfrac{9-3}{4-1} = 2$
5. With $y = 2x + 5$, calculate y for $x = 7$. 19

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: construct a scatter plot and choose the scales.

Activity 1. The table (x_i, y_i) gives the advertising spending and the sales over six months: $(1,12), (2,15), (3,17), (4,22), (5,24), (6,28)$. Choose scales, draw the scatter plot and plot the mean point.



Solution. $\bar{x} = 3.5$ and $\bar{y} = 19.67$ approximately. The scatter plot shows an increasing, roughly linear trend. Have students verbalise the piece of data that triggers the choice of method.

Written record. A scatter plot represents the pairs (x_i, y_i). The axes, units and scales must be indicated; each pair gives a single point.

Example 1. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $\bar{x} = 3.5$ and $\bar{y} = 19.67$ approximately. The scatter plot shows an increasing, roughly linear trend.

Exercise 5.1 · **essential.**

1. For $y = 3x + 13$, estimate y when $x = 8$.
2. Solve $3x + 13 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Solution. For $x = 8$, $y = 37$. The equation gives $x = \dfrac{37}{3}$. $x = 8$ gives an interpolation; $x = 30$ an extrapolation.

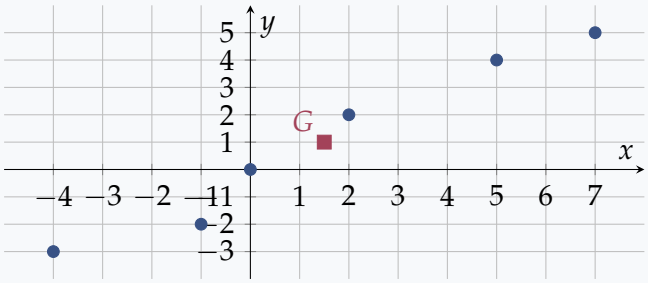
Application. Over six months, a convenience store compares the gap x_i between the month's average temperature and the seasonal norm (in °C) with the change y_i in its cold-drink sales compared with the same month of the previous year (in hundreds of bottles).

Temperature gap x_i (°C)	-4	-1	0	2	5	7
Change in sales y_i (hundreds)	-3	-2	0	2	4	5

1. State the variable placed on the horizontal axis, the one placed on the vertical axis, and their units.
2. Choose suitable scales, then plot the scatter plot of the points $M_i(x_i, y_i)$ on a coordinate grid.
3. Calculate the coordinates of the mean point G of the scatter plot and plot it.
4. Describe the shape of the scatter plot in one sentence.

Solution. **Target:** Building a scatter plot.

1. Horizontal axis: the temperature gap x, in °C; vertical axis: the change in sales y, in hundreds of bottles. Both variables take negative values: the axes must cross inside the graph.
2. One can take 1 cm for 1 °C (from -5 to 8) and 1 cm for one hundred bottles (from -4 to 6). Plot the six points $(-4, -3)$, $(-1, -2)$, $(0, 0)$, $(2, 2)$, $(5, 4)$ and $(7, 5)$, without joining them.



3. $\bar{x} = \dfrac{-4-1+0+2+5+7}{6} = \dfrac{9}{6} = 1.5$ and $\bar{y} = \dfrac{-3-2+0+2+4+5}{6} = \dfrac{6}{6} = 1$. Hence $G(1.5, 1)$.
4. The scatter plot rises: the warmer the month compared with the norm, the more cold-drink sales increase; the points lie close to a straight line.

Homework. Redo practice exercise 2 with the following data: replace the initial value with 275 and the rate or ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 32 — **Reading and interpreting a scatter plot**

Bivariate statistics — 28 Jan 2027 • B: 1 h – C: 2 h

Objectives.

- Describe a trend.
- Spot spread and outliers.
- Avoid unwarranted causality.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 7 - x + 2$. $2x + 9$
2. Calculate the mean of 7, 9 and 11. 9
3. Give the slope between $(1, 3)$ and $(4, 9)$. $\dfrac{9-3}{4-1} = 2$
4. With $y = 2x + 5$, calculate y for $x = 7$. 19
5. Solve $3x + 4 = 25$. $x = 7$

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: describe a trend and spot spread and outliers.

Activity 2. Compare three scatter plots: a tight increasing trend, a scattered decreasing trend, no trend. For each, write two sentences without using the word “cause”.

Solution. The description covers direction, shape, spread and outliers. A scatter plot alone does not establish causality. Have students verbalise the piece of data that triggers the choice of method.

Written record. Describing a scatter plot means commenting on its direction, shape, spread and outliers, without claiming a causality that has not been established.

Example 2. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

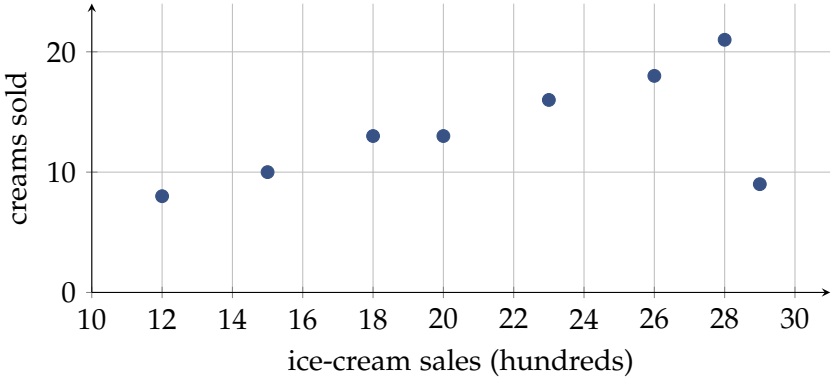
Solution. The description covers direction, shape, spread and outliers. A scatter plot alone does not establish causality.

Exercise 5.2 · essential.

1. For $y = 4x + 14$, estimate y when $x = 8$.
2. Solve $4x + 14 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Solution. For $x = 8$, $y = 46$. The equation gives $x = \dfrac{36}{4}$. $x = 8$ gives an interpolation; $x = 30$ an extrapolation.

Application. For eight summer weeks, a seaside ice-cream shop records its weekly sales x (in hundreds of cones); the neighbouring pharmacy records the number y of soothing sunburn creams sold in the same week. The resulting scatter plot is shown below.



1. Describe the scatter plot: direction, shape and spread.
2. One point stands clearly apart from the others. Give its coordinates and suggest a plausible explanation.
3. The pharmacist claims: “Ice creams cause sunburn.” What do you think of this claim?
4. Suggest a factor that could explain why the two series change together.

Solution. **Target:** Interpreting a scatter plot without inferring causation.

1. Apart from one point, the scatter plot is increasing: the weeks when the shop sells more ice cream are also the weeks when the pharmacy sells more creams. The points are fairly close to a straight line, with little spread.
2. The point $(29, 9)$ is an outlier: very high ice-cream sales but few creams sold. Possible explanation: a cloudy week with a festival that attracted many visitors without strong sun exposure.
3. The scatter plot only shows that the two variables move in the same direction (correlation). It does not prove that one variable causes the other: the claim is not justified by these data.
4. Sunshine (or heat, or tourist numbers) increases both ice-cream sales and sunburn: it is a factor common to both series.

Homework. Redo practice exercise 3 with the following data: replace the initial value with 280 and the rate or ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 33 — Linear regression

Bivariate statistics

2 Feb 2027 • B: 2 h – C: 1 h

Objectives.

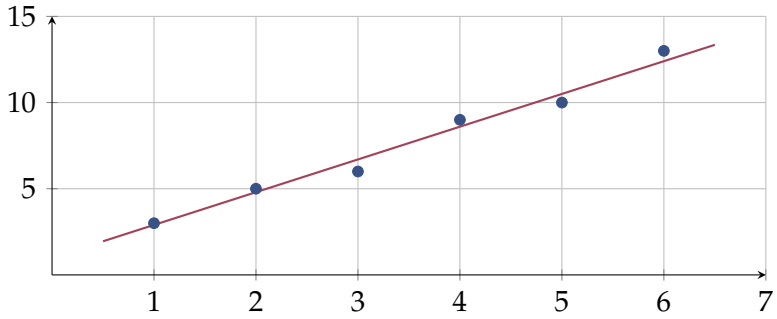
- Obtain a linear regression.
- Interpret a and b.
- Check with the mean point.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 20% of 160. 32
2. Give the slope between $(1,3)$ and $(4,9)$. $\frac{9-3}{4-1} = 2$
3. With $y = 2x + 5$, calculate y for $x = 7$. 19
4. Solve $3x + 4 = 25$. $x = 7$
5. Calculate the mean of 8, 10 and 12. 10

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: obtain a linear regression and interpret a and b.

Activity 3. With the data of session 31, obtain the linear regression with a calculator. Interpret the slope and check that the mean point lies approximately on the line.



Solution. A regression gives approximately $y = 3.17x + 8.57$. An extra hundred euros is associated with about 317 extra sales if y is in hundreds. Have students verbalise the piece of data that triggers the choice of method.

Written record. A line $y = ax + b$ fits a scatter plot. a represents the estimated average change in y for one extra unit of x.

Example 3. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. A regression gives approximately $y = 3.17x + 8.57$. An extra hundred euros is associated with about 317 extra sales if y is in hundreds.

Exercise 5.3 · essential.

1. For $y = 5x + 15$, estimate y when $x = 8$.
2. Solve $5x + 15 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Solution. For $x = 8$, $y = 55$. The equation gives $x = \dfrac{35}{5}$. $x = 8$ gives an interpolation; $x = 30$ an extrapolation.

Application. The manager of a gas-heated shop records, on some winter days, the average outdoor temperature x_i (in °C) and the shop's gas consumption y_i (in kWh).

x_i (°C)	-4	-2	0	3	6	9
y_i (kWh)	62	55	50	41	32	24

1. Plot the scatter plot on a coordinate grid, then calculate the coordinates of the mean point G.
2. An intern suggests the line d_1 with equation $y = -2x + 50$. Draw it. Does this fit seem satisfactory? Justify using the point G.
3. From now on, the line d_2 with equation $y = -3x + 50$ is used. Check that G lies on d_2 and interpret the numbers -3 and 50 in context.
4. Using d_2, estimate:
   - the consumption on a day when it is 1 °C;
   - the consumption on a day when it is 12 °C;
   - the temperature on a day when the consumption is 35 kWh.

   For the first two estimates, state whether it is an interpolation or an extrapolation.

Solution. **Target:** Assessing and using a fitted line.

1. Plot the six points. $\bar{x} = \dfrac{-4-2+0+3+6+9}{6} = \dfrac{12}{6} = 2$ and $\bar{y} = \dfrac{62+55+50+41+32+24}{6} = \dfrac{264}{6} = 44$, so $G(2, 44)$. The scatter plot is decreasing and almost aligned.
2. For $x = 2$, d_1 gives $y = -2 \times 2 + 50 = 46 \neq 44$: d_1 does not pass through G. Moreover, it is not steep enough: it runs below the first points (58 instead of 62 at -4 °C) and well above the last ones (32 instead of 24 at 9 °C). The fit is not satisfactory.
3. $-3 \times 2 + 50 = 44$: $G \in d_2$. The coefficient -3 means that one more degree corresponds, on

average, to about 3 kWh less consumed; 50 is the estimated consumption on a day at 0 °C.

4. For $x = 1$: $y = -3 + 50 = 47$, i.e. about 47 kWh (interpolation, since $1 \in [-4, 9]$).
   For $x = 12$: $y = -36 + 50 = 14$, i.e. about 14 kWh (extrapolation, since $12 > 9$: to be interpreted with caution).
   $-3x + 50 = 35$ is equivalent to $x = 5$: a consumption of 35 kWh corresponds to about 5 °C.

Homework. Redo practice exercise 2 with the following data: replace the initial value with 285 and the rate or ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 34 — Interpolation

Bivariate statistics

4 Feb 2027 • B: 1 h – C: 2 h

Objectives.

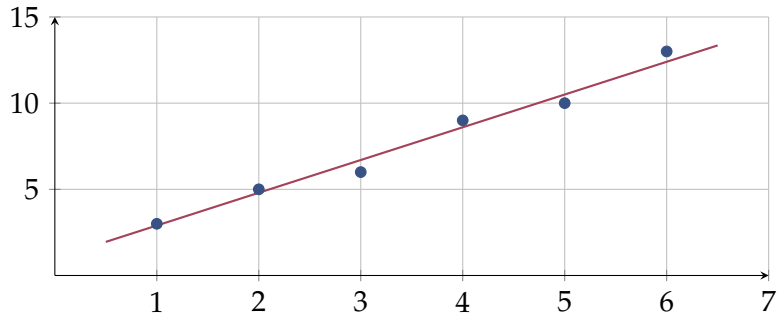
- Interpolate.
- Solve using the model.
- Express an estimate.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 25% increase. $1 + \dfrac{25}{100} = 1.25$
2. With $y = 2x + 5$, calculate y for $x = 7$. 19
3. Solve $3x + 4 = 25$. $x = 7$
4. Calculate the mean of 9, 11 and 13. 11
5. Give the slope between $(1, 3)$ and $(4, 9)$. $\dfrac{9-3}{4-1} = 2$

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: interpolate and solve using the model.

Activity 4. Over the observed period $x \in [1, 12]$, the model is $y = 2.8x + 14$. Estimate y for $x = 7.5$, then estimate x when $y = 35$. Classify each estimate.



Solution. $y = 35$ and $x = 7.5$. Both values lie inside the observed range: interpolations. Have students verbalise the piece of data that triggers the choice of method.

Written record. An interpolation estimates a value inside the observed range. The result is an estimate that depends on the model.

Example 4. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $y = 35$ and $x = 7.5$. Both values lie inside the observed range: interpolations.

Exercise 5.4 · essential.

1. For $y = 6x + 16$, estimate y when $x = 8$.
2. Solve $6x + 16 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Solution. For $x = 8$, $y = 64$. The equation gives $x = \dfrac{34}{6}$. $x = 8$ gives an interpolation; $x = 30$ an extrapolation.

Application. A regional operator records the number of subscribers to its fibre-optic offer on 31 December each year.

Year	2018	2019	2020	2021	2022	2023
Year rank x_i	0	1	2	3	4	5
Subscribers y_i (thousands)	8.2	11.5	13.9	17.4	20.1	23.8

1. Plot the scatter plot of the points $M_i(x_i, y_i)$. Is a linear fit reasonable?
2. Using a calculator, find an equation of the least-squares regression line of y on x. Round the coefficients to 10^{-2}.
3. Assuming the trend continues, estimate the number of subscribers at the end of 2026.
4. According to this model, in which year would the number of subscribers exceed 40,000?

Solution. **Target:** Linear regression and time-series forecast.

1. The six points are almost aligned along an increasing straight line: a linear fit is relevant.
2. The calculator gives $a \approx 3.0657$ and $b \approx 8.1524$, hence the line $y = 3.07x + 8.15$.
3. The end of 2026 corresponds to rank $x = 8$: $y = 3.07 \times 8 + 8.15 = 32.71$. About 32,710 subscribers can be forecast. This is an extrapolation: the result assumes that the trend observed from 2018 to 2023 continues.
4. Solve $3.07x + 8.15 > 40$, i.e. $x > \dfrac{40 - 8.15}{3.07} \approx 10.37$. The first suitable whole rank is $x = 11$ (check: $3.07 \times 10 + 8.15 = 38.85 < 40$ and $3.07 \times 11 + 8.15 = 41.92 > 40$). The threshold would be exceeded in $2018 + 11 = 2029$.

Homework. Redo practice exercise 3 with the following data: replace the initial value with 290 and the rate or ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 35 — Extrapolation

Bivariate statistics

9 Feb 2027 • B: 2 h – C: 1 h

Objectives.

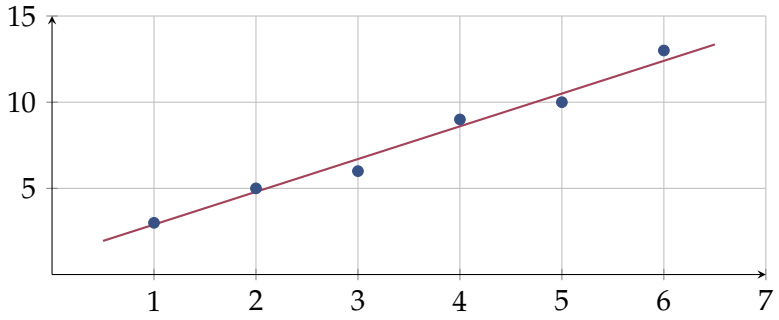
- Extrapolate with caution.
- Assess plausibility.
- Present a limitation of the model.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 3 - x + 2$. $2x + 5$
2. Solve $3x + 4 = 25$. $x = 7$
3. Calculate the mean of 3, 5 and 7. 5
4. Give the slope between $(1, 3)$ and $(4, 9)$. $\dfrac{9-3}{4-1} = 2$
5. With $y = 2x + 5$, calculate y for $x = 7$. 19

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: extrapolate with caution and assess plausibility.

Activity 5. The same model $y = 2.8x + 14$, fitted on $[1, 12]$, is used for $x = 40$. Calculate the estimate, then give three reasons why it may be unreliable.



Solution. The estimate is 126. It is far from the data, the trend may change and real-world constraints may cap the variable. Have students verbalise the piece of data that triggers the choice of method.

Written record. An extrapolation estimates outside the observed range. The further away we go, the more caution is needed.

Example 5. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. The estimate is 126. It is far from the data, the trend may change and real-world constraints may cap the variable.

Exercise 5.5 · essential.

1. For $y = 2x + 12$, estimate y when $x = 8$.
2. Solve $2x + 12 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Solution. For $x = 8$, $y = 28$. The equation gives $x = \dfrac{38}{2}$. $x = 8$ gives an interpolation; $x = 30$ an extrapolation.

Application. A logistics hub recorded the number of parcels shipped each year (in thousands).

Year	2019	2020	2021	2022	2023
Year rank x_i	0	1	2	3	4
Parcels shipped y_i (thousands)	212.4	268.0	341.7	398.2	463.6

1. Plot the scatter plot of the points $M_1, \ldots, M_5$ on an orthogonal grid.
2. The scatter plot is fitted by the line (M_1M_5) through the first and last points. Find an equation of this line (coefficients rounded to one decimal place if necessary).
3. Using this model, estimate the number of parcels shipped in 2027.
4. According to this model, in which year would the number of parcels shipped exceed one million?
5. Give two reasons why these forecasts may turn out to be unreliable.

Solution. **Target:** Line through two points and careful extrapolation.

1. The five points are almost aligned along an increasing straight line.
2. $M_1(0, 212.4)$ and $M_5(4, 463.6)$. The gradient is $\dfrac{463.6 - 212.4}{4 - 0} = \dfrac{251.2}{4} = 62.8$ and the intercept is 212.4 (abscissa 0). Equation: $y = 62.8x + 212.4$.
3. 2027 corresponds to rank $x = 8$: $y = 62.8 \times 8 + 212.4 = 714.8$, i.e. about 714,800 parcels.
4. One million parcels corresponds to $y = 1{,}000$. Solve $62.8x + 212.4 > 1{,}000$, i.e. $x > \dfrac{787.6}{62.8} \approx 12.54$. The first whole rank is $x = 13$ (check: $62.8 \times 12 + 212.4 = 966 < 1{,}000$ and $62.8 \times 13 + 212.4 = 1{,}028.8$). The threshold would be crossed in $2019 + 13 = 2032$.
5. These are extrapolations far from the observed period (2019–2023): the trend may change (competition, economic crisis, the hub reaching full capacity); moreover the line is built from only two points.

Homework. Redo practice exercise 2 with the following data: replace the initial value with 295 and the rate or ratio with 7. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 36 — **Change of variable**

Bivariate statistics — 11 Feb 2027 • B: 1 h – C: 2 h

Objectives.

- Carry out a change of variable.
- Fit the new data.
- Return to the original variable.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 5% of 80. 4
2. Calculate the mean of 4, 6 and 8. 6
3. Give the slope between $(1,3)$ and $(4,9)$. $\frac{9-3}{4-1}=2$
4. With $y=2x+5$, calculate y for $x=7$. 19
5. Solve $3x+4=25$. $x=7$

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: carry out a change of variable and fit the new data.

Activity 6. The data appear to be exponential. We set $z=\log(y)$ and the fit gives $z=0.08x+1.2$. Express y as a function of x, then calculate y for $x=5$.

Solution. $y=10^{0.08x+1.2}$. For $x=5$, $y=10^{1.6}\approx 39.8$. Have students verbalise the piece of data that triggers the choice of method.

Written record. A change of variable can make a scatter plot linear. After fitting in the new variable, we must return to the original variable.

Example 6. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $y=10^{0.08x+1.2}$. For $x=5$, $y=10^{1.6}\approx 39.8$.

Exercise 5.6 · essential.

1. For $y=3x+13$, estimate y when $x=8$.
2. Solve $3x+13=50$.
3. The model was built on $[2,12]$: classify the estimates for $x=8$ and $x=30$.

Solution. For $x = 8$, $y = 37$. The equation gives $x = \dfrac{37}{3}$. $x = 8$ gives an interpolation; $x = 30$ an extrapolation.

Application. A marketing agency studies, for several clients, the monthly online advertising budget x_i (in hundreds of euros) and the monthly number of visits y_i to the client's website (in thousands).

x_i	1	2	4	5	8	10	15	20	30	40
y_i	13	19	28	29	35	36	42	44	49	52

Visits increase more and more slowly as the budget increases: a linear fit of y on x is not relevant. Let $u_i = \log(x_i)$.

1. Calculate the values u_i, rounded to two decimal places.
2. Plot the scatter plot of the points (u_i, y_i). Using a calculator, find an equation of the least-squares regression line of y on u (coefficients rounded to one decimal place).
3. Deduce an expression of y in terms of x.
4. Estimate the number of visits for a budget of 1,200 €, then of 6,000 €. Comment on the reliability of each estimate.

Solution. **Target:** Linearising with a logarithmic change of variable.

1. u_i: 0; 0.30; 0.60; 0.70; 0.90; 1; 1.18; 1.30; 1.48; 1.60.
2. The scatter plot of (u_i, y_i) is almost aligned. The calculator gives (with the rounded u_i) $a \approx 24.59$ and $b \approx 12.42$, hence $y = 24.6u + 12.4$.
3. Since $u = \log(x)$, we get $y = 24.6\log(x) + 12.4$.
4. 1,200 € corresponds to $x = 12$: $y = 24.6\log(12) + 12.4 \approx 38.9$, i.e. about 38,900 visits (interpolation, $12 \in [1, 40]$).
   6,000 € corresponds to $x = 60$: $y = 24.6\log(60) + 12.4 \approx 56.1$, i.e. about 56,100 visits. This is an extrapolation ($60 > 40$): the estimate should be interpreted with caution.

Homework. Redo practice exercise 3 with the following data: replace the initial value with 300 and the rate or ratio with 2. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Notes

Session 37 — Choosing the model and fitting

Bivariate statistics

2 Mar 2027 • B: 2 h – C: 1 h

Objectives.

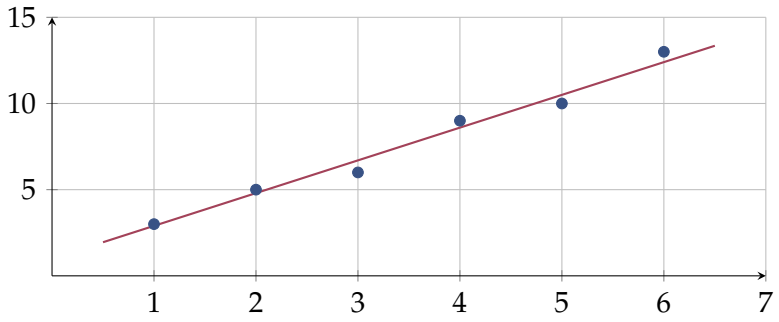
- Compare models.
- Use residuals.
- Define a range of use.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 10% increase. $1 + \dfrac{10}{100} = 1.10$
2. Give the slope between $(1,3)$ and $(4,9)$. $\dfrac{9-3}{4-1} = 2$
3. With $y = 2x + 5$, calculate y for $x = 7$. 19
4. Solve $3x + 4 = 25$. $x = 7$
5. Calculate the mean of 5, 7 and 9. 7

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: compare models and use residuals.

Activity 7. Two models give total absolute residuals of 18.4 and 11.7 over the observed range. The second, however, shows unrealistic growth beyond $x = 20$. Choose a model to interpolate at $x = 12$, then discuss extrapolation at $x = 50$.



Solution. The second is preferable for interpolating over the observed range if the other criteria are comparable; no automatic choice is justified for $x = 50$. Have students verbalise the piece of data that triggers the choice of method.

Written record. We choose a model according to the shape of the scatter plot, the size of the residuals and the range of use. A better visual fit does not allow unlimited extrapolation.

Example 7. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. The second is preferable for interpolating over the observed range if the other criteria are comparable; no automatic choice is justified for $x = 50$.

Exercise 5.7 · essential.

1. For $y = 4x + 14$, estimate y when $x = 8$.
2. Solve $4x + 14 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Solution. For $x = 8$, $y = 46$. The equation gives $x = \dfrac{36}{4}$. $x = 8$ gives an interpolation; $x = 30$ an extrapolation.

Application. A start-up is calibrating a low-cost carbon dioxide (CO_2) sensor intended for offices. Placed in chambers with a known concentration C (in ppm, parts per million), the sensor delivers a voltage U (in volts).

Voltage U (V)	0.42	0.81	1.23	1.60	2.05	2.41
Concentration C (ppm)	400	800	1,200	1,600	2,000	2,400

1. Plot the scatter plot of the points (U, C), with U on the horizontal axis.
2. Does a linear fit seem relevant? Using a calculator, find an equation of the regression line of C on U, in the form $C = aU + b$ (coefficients rounded to the nearest unit).
3. Calculate the residual for $U = 1.23$, that is, the difference between the actual concentration and the one given by the model. Comment.
4. In an office, the sensor reads 1.4 V. Estimate the CO_2 concentration. A threshold of 1,000 ppm is often used as a guide for airing a room: should the room be aired?
5. Can this model be used with the same confidence for a voltage of 3.5 V? Justify.

Solution. **Target:** Calibration: fit, residual and validity range.

1. The six points are almost aligned along an increasing straight line.
2. Yes. The calculator gives $a \approx 996.59$ and $b \approx -15.16$, hence $C = 997U - 15$.
3. The model gives $997 \times 1.23 - 15 = 1{,}211.31$. The residual is $1{,}200 - 1{,}211.31 \approx -11$ ppm, i.e. less than 1% of the value: the fit is good.
4. $C = 997 \times 1.4 - 15 = 1{,}380.8$, i.e. about 1,381 ppm (interpolation, since $1.4 \in [0.42, 2.41]$). This value exceeds 1,000 ppm: the room should be aired.
5. The model gives $997 \times 3.5 - 15 = 3{,}474.5$ ppm, but 3.5 V lies outside the calibration interval $[0.42, 2.41]$: this is an extrapolation. Nothing guarantees that the sensor stays linear at high concentrations; the calibration should be extended.

Homework. Redo practice exercise 2 with the following data: replace the initial value with 305 and the rate or ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 38 — **Review and assessment**

Bivariate statistics — 4 Mar 2027 • B: 1 h – C: 2 h

Objectives.

- Choose and use a regression.
- Write a statistical conclusion.
- Analyse an error.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 6 - x + 2$. $2x + 8$
2. With $y = 2x + 5$, calculate y for $x = 7$. 19
3. Solve $3x + 4 = 25$. $x = 7$
4. Calculate the mean of 6, 8 and 10. 8
5. Give the slope between $(1,3)$ and $(4,9)$. $\dfrac{9-3}{4-1} = 2$

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: choose and use a regression and write a statistical conclusion.

Activity 8. Starting from a scatter plot supplied by the teacher, produce a complete answer: choice of model, equation, estimate, type of estimate, unit and caveat. Swap papers and use a six-criterion grid.

Solution. The marking checks the consistency of the model and the presence of the six required elements, not only the numerical result. Have students verbalise the piece of data that triggers the choice of method.

Written record. A complete statistical answer contains the model, the calculation, the rounding, the unit, the type of estimate and a caveat on validity.

Example 8. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. The marking checks the consistency of the model and the presence of the six required elements, not only the numerical result.

Exercise 5.8 · essential.

1. For $y = 5x + 15$, estimate y when $x = 8$.
2. Solve $5x + 15 = 50$.

3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Solution. For $x = 8$, $y = 55$. The equation gives $x = \dfrac{35}{5}$. $x = 8$ gives an interpolation; $x = 30$ an extrapolation.

Application. For each question, exactly one option is correct. Choose it and justify.
Three stores of the same chain have a sales area x (in hundreds of m^2) and a monthly turnover y (in tens of thousands of euros) represented by the points $M_1(2, 5)$, $M_2(4, 11)$ and $M_3(6, 11)$.

1. The mean point G of this scatter plot is:
   a. the point M_2;
   b. the point with coordinates $(4, 9)$;
   c. the point with coordinates $(12, 27)$.
2. An equation of the line (M_1M_3) is:
   a. $y = 1.5x + 2$;
   b. $y = 2x + 1$;
   c. $y = 1.5x + 3$.
3. The least-squares regression line obtained with a calculator:
   a. passes through the point M_2;
   b. passes through G and has the same gradient as (M_1M_3);
   c. passes through the origin.
4. Using this regression line, the estimated turnover of a 500 m^2 store is:
   a. 95,000 € ;
   b. 105,000 € ;
   c. 110,000 €.

Solution. **Target:** Mean point and regression line: justified choices.

1. Answer b: $\bar{x} = \dfrac{2+4+6}{3} = 4$ and $\bar{y} = \dfrac{5+11+11}{3} = 9$. Option c forgets to divide the sums by 3.
2. Answer a: the gradient is $\dfrac{11-5}{6-2} = 1.5$ and $5 = 1.5 \times 2 + b$ gives $b = 2$. The line $y = 2x + 1$ passes through M_1 but not through M_3 ($2 \times 6 + 1 = 13 \neq 11$).
3. Answer b: the calculator gives $y = 1.5x + 3$. It passes through G ($1.5 \times 4 + 3 = 9$), has gradient 1.5 like (M_1M_3), and passes neither through M_2 ($1.5 \times 4 + 3 = 9 \neq 11$) nor through the origin ($b = 3 \neq 0$).
4. Answer b: 500 m^2 corresponds to $x = 5$, and $y = 1.5 \times 5 + 3 = 10.5$ tens of thousands of euros, i.e. 105,000 € (interpolation).

Homework. Redo practice exercise 3 with the following data: replace the initial value with 310 and the rate or ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.